

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from ydata_profiling import ProfileReport
import sweetviz as sv
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
```

02 CFPB Exploratory Data Analysis (EDA)

Objective: profile the trimmed CFPB credit card complaint subset, surface the dominant issue and firm patterns within that portfolio, and document region-based differences that motivate the time-series analysis in notebooks 03-04.

```
In [ ]: # Load trimmed credit card subset data
df = pd.read_parquet("../data/processed/credit_card_dataset.parquet")
```

```
In [3]: print(f"Loaded {len(df):,} rows & {len(df.columns)} columns")

df.head()
```

Loaded 482,799 rows & 22 columns

Out[3]:

	Date received	Product	Sub-product	Issue	Sub-issue	Consumer complaint narrative	Company public response	
0	2020-03-09	Credit card or prepaid card	General-purpose credit card or charge card	Fees or interest	Problem with fees	On XX/XX/2020 I booked a hotel in XXXX that re...	None	JCF
1	2020-03-18	Credit card or prepaid card	General-purpose credit card or charge card	Fees or interest	Problem with fees	I have had a Barclays/Uber credit card for nea...	Company has responded to the consumer and the ...	
2	2023-12-15	Credit card	General-purpose credit card or charge card	Problem with a company's investigation into an...	Was not notified of investigation status or re...	Take a look at the attached documents. I reque...	Company has responded to the consumer and the ...	TR/INTI
3	2020-01-04	Credit card or prepaid card	Government benefit card	Trouble using the card	Trouble getting information about the card	I received my new card for my child support an...	Company has responded to the consumer and the ...	AS:
4	2019-07-31	Credit card or prepaid card	General-purpose credit card or charge card	Fees or interest	Problem with fees	None	Company has responded to the consumer and the ...	CITI

5 rows × 22 columns

The trimmed credit card dataset loaded successfully and is ready for portfolio-specific profiling.

1. Global EDA Profiling

This section profiles the credit card complaint subset only. With product scope fixed by construction, the main goals are to summarize issue mix, firm concentration, response-process fields, and time-based growth patterns before moving to region-level comparisons.

```
In [11]: profile = ProfileReport(  
         df,
```

```

title="CFPB Credit Card Complaints - YData Profiling Report",
explorative=True # richer, but still reasonable runtime
)

```

```

In [12]: try:
          profile.to_notebook_iframe()
        except MemoryError:
          print("MemoryError on profile render; using a 100k-row fallback for notebook di
profile_sample = ProfileReport(
    df.sample(n=min(len(df), 100_000), random_state=42),
    title="CFPB Credit Card Complaints Profile (100k-row fallback)",
    explorative=True,
    minimal=True,
)
          profile_sample.to_notebook_iframe()

```

Summarize dataset: 0%| | 0/5 [00:00<?, ?it/s]

100%|██████████| 22/22 [01:23<00:00, 3.80s/it]

Generate report structure: 0%| | 0/1 [00:00<?, ?it/s]

Render HTML: 0%| | 0/1 [00:00<?, ?it/s]



Overview

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Dataset statistics

Number of variables	22
Number of observations	482799
Missing cells	1664593
Missing cells (%)	15.7%
Duplicate rows	0
Duplicate rows (%)	0.0%
Total size in memory	859.1 MiB
Average record size in memory	1.8 KiB

Variable types

DateTime	3
Categorical	8
Text	8
Boolean	2
Numeric	1

Variables

```
In [13]: # Save to HTML
profile.to_file("../reports/cfpb_credit_card_profile.html")
```

Export report to file: 0%| | 0/1 [00:00<?, ?it/s]

1.1 Key Univariate Insights

Because this notebook now focuses only on credit card complaints, the profiling output should be interpreted as a within-portfolio summary rather than a comparison across CFPB products.

Summary highlights:

- Product composition is fixed to the credit card portfolio, so the most informative univariate patterns come from issue categories, complaint narratives, firm concentration, and operational response fields.
- Complaint volume remains concentrated in a relatively small number of recurring issue themes and large firms.
- Response-process fields still provide useful operational context, even though the analysis is now narrower than the original all-product workflow.
- The date distribution remains right-skewed, which is consistent with stronger complaint volume in the more recent years of the trimmed sample.

These patterns motivate the regional comparison below and provide the descriptive baseline for the credit card forecasting notebooks.

1.2 South vs Others Segment Comparison

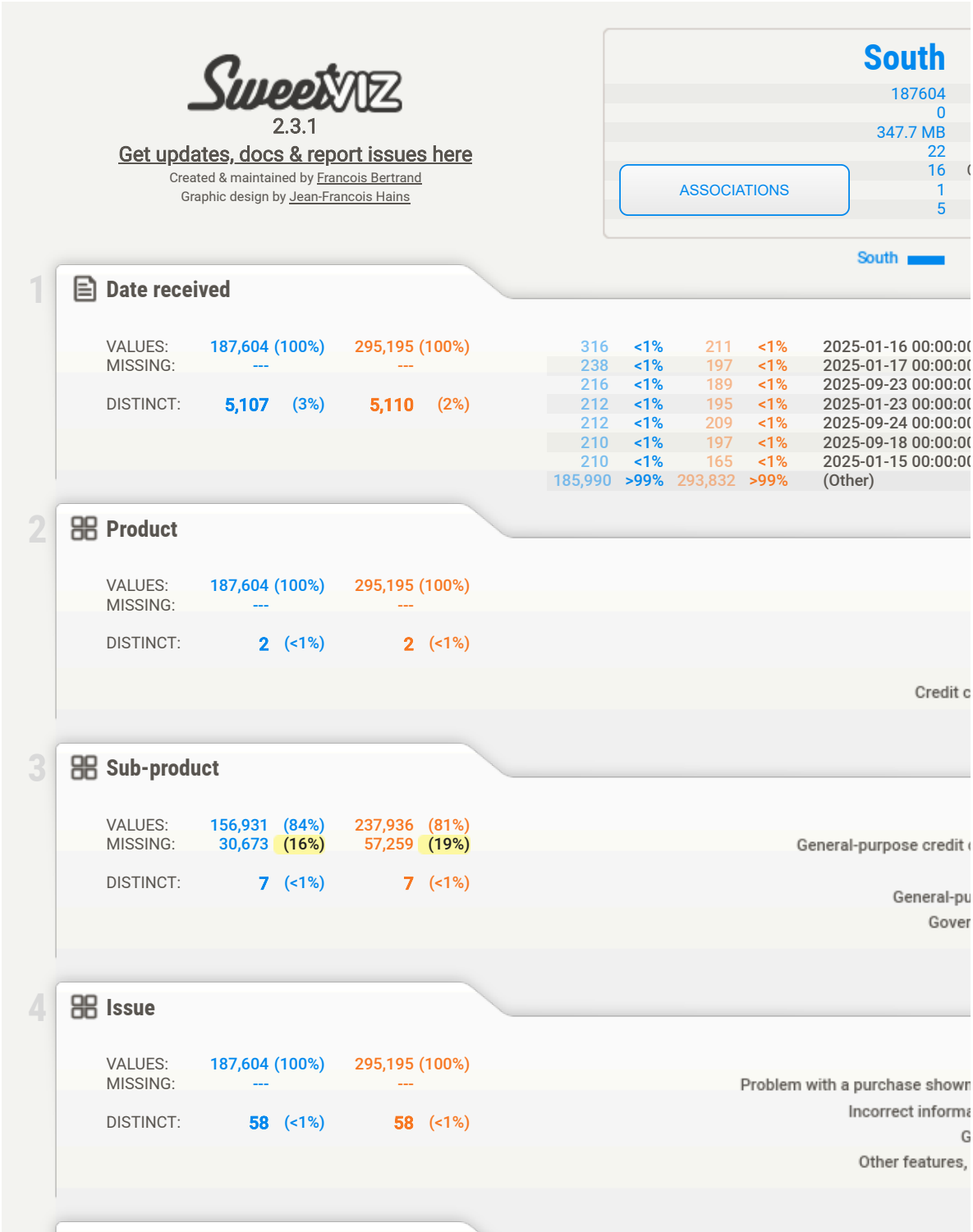
Given the project focus on regional heterogeneity, this section compares South against all other regions within the credit card subset to identify differences in complaint composition, firm exposure, and operational response characteristics.

```
In [ ]: # Split the credit card subset into South vs non-South regions
south_df = df[df['region'] == 'South'].copy()
other_df = df[df['region'] != 'South'].copy()
```

```
In [8]: # Generate comparison report
sweet_report = sv.compare([south_df, "South"], [other_df, "Others"])

|          | [ 0%]  00:00 -> (? left)
```

```
In [9]: # Notebook iframe
sweet_report.show_notebook(scale=0.9)
```



```
In [10]: sweet_report.show_html('../reports/south_vs_others_sweetviz.html')
```

Report ../reports/south_vs_others_sweetviz.html was generated! NOTEBOOK/COLAB USERS: the web browser MAY not pop up, regardless, the report IS saved in your notebook/colab files.

South vs Others: Review Summary

Within the credit card portfolio, the South segment still shows meaningful differences from the combined non-South regions, and those differences are relevant for downstream modeling.

Key takeaways:

- The regional comparison is now about variation inside the credit card subset, not about differences across CFPB product lines.
- South and non-South records can differ in issue mix, firm exposure, and geographic concentration, which is enough to justify checking regional behavior separately.
- Operational response fields remain broadly comparable, but directional differences are still useful when interpreting segment-level forecasting results.

Implications for notebooks 03-04:

- The region split remains analytically useful even after narrowing the project to credit card complaints.
- South-vs-others evaluation is still justified as a descriptive segmentation strategy for the forecasting workflow.
- Findings should still be treated as descriptive evidence from the prepared credit card subset, not as causal claims.